

# Copper Electroplating

## Experiment Packet

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**Audience:** instructors and teaching assistants

**Purpose:** Bench instructions for instructors and TAs

**Session hook:** Students visibly transform a metal object

**Length:** 90-minute lab block (~20 min concepts + ~70 min hands-on)

### This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

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Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

## Session 2 — Copper Electroplating

**Hook:** Students visibly transform a metal object.

### Learning outcomes

- Explain electrolysis and distinguish it from a galvanic cell
- Identify cathode and anode in an **electrolytic** cell
- Describe how current controls deposition rate
- Explain why surface preparation and current density affect deposit quality

### Session plan (90 min)

| Time      | Activity                                                                  | Reference                 |
|-----------|---------------------------------------------------------------------------|---------------------------|
| 0–10 min  | Show blue $\text{CuSO}_4$ solution / dissolving crystals; preview plating | lecture.md — Hook         |
| 10–25 min | Concept: ion deposition at cathode, anode dissolution                     | lecture.md                |
| 25–35 min | Clean cathode object (sand, rinse, no fingerprints)                       | experiment.md — Prep      |
| 35–50 min | Set up plating cell                                                       | experiment.md — Setup     |
| 50–70 min | Plate 2–10 min; observe every 1–2 min                                     | experiment.md — Plating   |
| 70–80 min | Compare variables                                                         | experiment.md — Variables |
| 80–90 min | Discussion: surface prep and current density                              | lecture.md — Wrap-up      |

### Key reactions

Cathode (–):  $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$

Anode (+):  $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^-$

### Files

- lecture.md — Concept block
- experiment.md — Plating procedure
- materials.md — Checklist
- preparation.md —  $\text{CuSO}_4$  solution prep

### Status

- [ ] 0.5 M  $\text{CuSO}_4$  mixed and labeled
- [ ] Plating procedure pilot-tested
- [ ] Lecture slides/diagrams ready

## Session 2 — Preparation

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### Before Session 2 (instructor)

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- ☐ Prepare **0.5 M CuSO<sub>4</sub>** plating bath (e.g. 62 g CuSO<sub>4</sub>·5H<sub>2</sub>O per 500 mL tap water); label concentration
- ☐ Pilot plate one nail: record time, current, appearance
- ☐ Pre-sand a few demo objects
- ☐ Source resistors if using current limiting

### Mixing 0.5 M CuSO<sub>4</sub> (quick recipe)

| Batch size | CuSO <sub>4</sub> ·5H <sub>2</sub> O (blue crystals) | Water          |
|------------|------------------------------------------------------|----------------|
| 200 mL     | <b>25 g</b>                                          | Fill to 200 mL |
| 500 mL     | <b>62 g</b>                                          | Fill to 500 mL |
| 1.0 L      | <b>125 g</b>                                         | Fill to 1.0 L  |

1. Weigh CuSO<sub>4</sub>·5H<sub>2</sub>O; dissolve in ~¾ of the water; stir until clear blue.
2. Top up to final volume; label "**0.5 M CuSO<sub>4</sub> — Session 2 — do not drink**".
3. Tap water is fine; use distilled only if hard tap water makes the bath milky.

### Day-of setup

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- ☐ Goggles and gloves at each station
- ☐ Aliquot **50–100 mL** CuSO<sub>4</sub> per group jar
- ☐ One 9 V battery per group (fresh if possible)
- ☐ Waste bucket labeled "copper waste — do not drain"
- ☐ Demo cell ready for hook (jar + electrodes + meter)

### Open questions / notes

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- CuSO<sub>4</sub> concentration mixed (target 0.5 M):
- Resistor value used in pilot:
- Best demo object (nail vs. key):

## Session 2 — Materials

### Per group

- ☐ **0.5 M CuSO<sub>4</sub>** plating solution (50–100 mL)
- ☐ Copper gasket, strip, or wire (anode)
- ☐ Steel nail, screw, coin, washer, or key (cathode)
- ☐ 9 V battery or low-voltage DC supply
- ☐ Alligator clips
- ☐ Multimeter
- ☐ Optional: 100–500  $\Omega$  resistor (target ~10–40 mA)
- ☐ Sandpaper or steel wool
- ☐ Jam jar or beaker (~100 mL)
- ☐ Gloves and goggles

### Instructor — 0.5 M CuSO<sub>4</sub> stock

| Batch  | CuSO <sub>4</sub> ·5H <sub>2</sub> O | Water     |
|--------|--------------------------------------|-----------|
| 200 mL | 25 g                                 | to 200 mL |
| 500 mL | 62 g                                 | to 500 mL |
| 1 L    | 125 g                                | to 1 L    |

- ☐ Extra copper anode material
- ☐ Waste container for copper-bearing liquids
- ☐ Paper towels for rinsing/drying

### Pilot notes

| Parameter                       | Target from pilot       |
|---------------------------------|-------------------------|
| Solution                        | 0.5 M CuSO <sub>4</sub> |
| Solution volume                 | ~50–100 mL              |
| Typical current (with resistor) | ___ mA                  |
| Good plating time               | ___ min                 |

## Session 2 — Experiment: Copper Electroplating

**Instructors:** polarity checks, deposit quality, and solution notes — instructor.md.

### Copper solution — what to use

You need a blue **copper(II)** plating bath from **copper sulfate**. Mix before class (or shortly before students plate). Do **not** rely on overnight vinegar/copper jars.

#### Recipe — 0.5 M $\text{CuSO}_4$ (recommended)

| Batch size              | $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue crystals) | Water          |
|-------------------------|-----------------------------------------------------------|----------------|
| 200 mL (about 2 groups) | <b>25 g</b>                                               | Fill to 200 mL |
| 500 mL                  | <b>62 g</b>                                               | Fill to 500 mL |
| 1.0 L                   | <b>125 g</b>                                              | Fill to 1.0 L  |

**Water:** tap water is fine. Distilled is slightly better if your tap water is very hard (bath may look milky). Save distilled water for Session 3 silver nitrate.

#### How to mix:

1. Weigh  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  (copper sulfate pentahydrate — the common blue lab salt).
2. Dissolve in  $\sim 3/4$  of the water; stir until clear blue.
3. Add water to the final volume; label "**0.5 M  $\text{CuSO}_4$  — Session 2 — do not drink**".
4. Store covered; goggles/gloves when pouring.

**Acceptable range if you must improvise:** 0.25–1.0 M. Below  $\sim 0.25$  M plating is slow; much above 1 M can give rough deposits and waste chemical.

**Do not add strong acid** (industrial baths sometimes use  $\text{H}_2\text{SO}_4$  — skip that for this age group).

#### Volume per group today

**50–100 mL** of 0.5 M  $\text{CuSO}_4$  in a jam jar or small beaker (enough to immerse both electrodes).

### Prep — Clean the cathode (25–35 min)

#### Object options

Steel nail, screw, coin, washer, or key (one “good” object + optional unplated control of the same type).

#### Procedure

1. Sand or steel-wool the surface until **bright metal** shows (no rust, paint, or dull oxide).
2. Rinse under water; pat dry with a paper towel.

3. **Do not touch** the area that will be plated — hold by the head, thread, or edge only. Finger oil ruins adhesion.
4. Optional: photograph or weigh before plating for comparison.

### Quality checklist

- [ ] No visible rust or grease
- [ ] Uniform bright sanded surface
- [ ] Rinsed and dry
- [ ] Placed on a clean paper towel until use

### Setup — Plating cell (35–50 min)

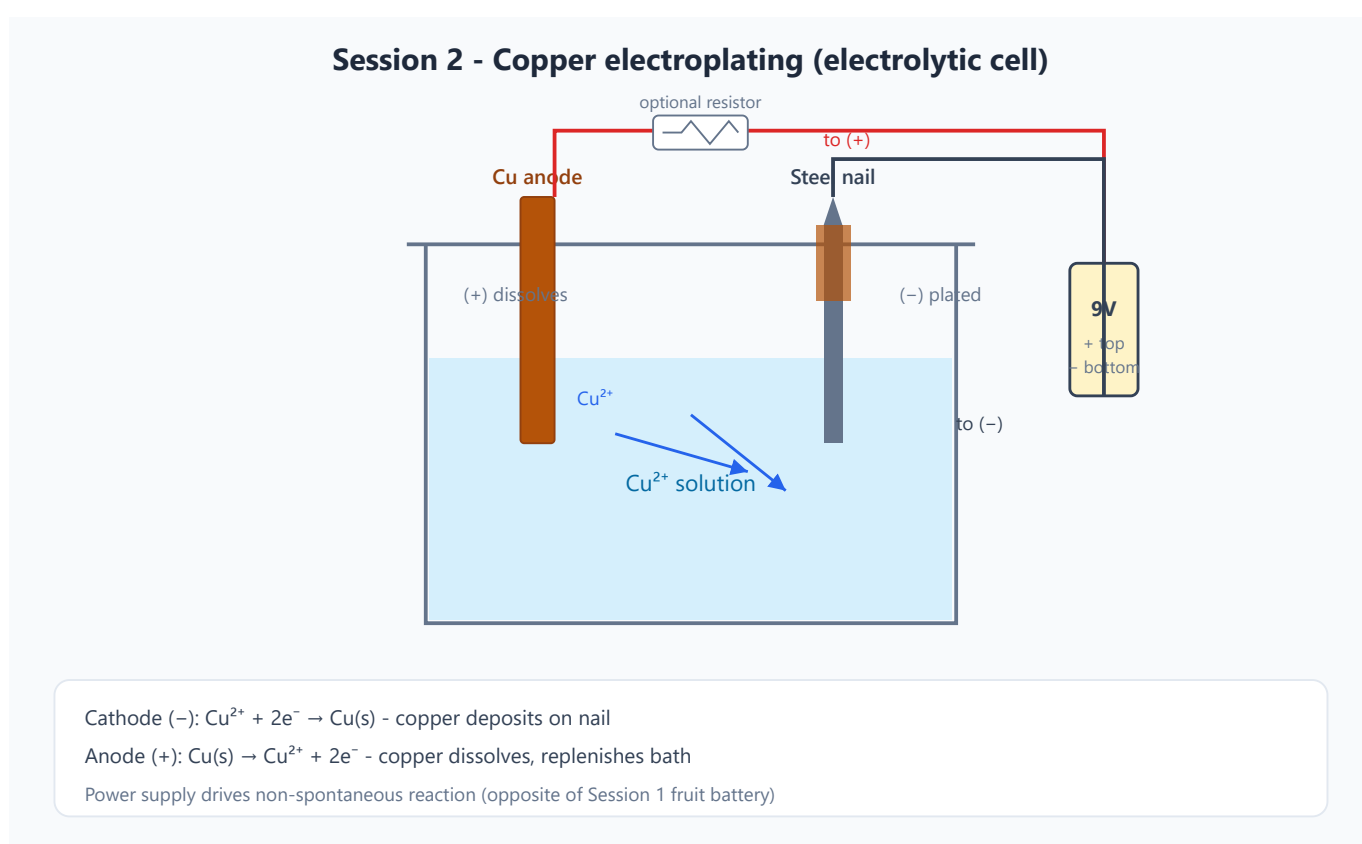


Figure 1 — Electrolytic cell: Cu anode (+) dissolves;  $\text{Cu}^{2+}$  deposits on the nail cathode (-). Optional resistor limits current.

### Materials at the station

- 50–100 mL **0.5 M  $\text{CuSO}_4$**
- Copper strip, gasket, or thick wire (**anode**)
- Cleaned object (**cathode**)
- 9 V battery (fresh)
- Alligator clips (at least 2)
- Optional: 100–500  $\Omega$  resistor, multimeter on DC mA

## Wiring — follow exactly

| Part                         | Connects to                               |
|------------------------------|-------------------------------------------|
| Object to plate (nail, etc.) | Battery <b>negative (-)</b> / black       |
| Copper anode                 | Battery <b>positive (+)</b> / red         |
| Optional resistor            | In series on either lead (limits current) |
| Optional ammeter             | In series on DC <b>mA</b> range           |

**Memory aid:** *The object that should gain copper goes to the minus sign.*

## Procedure

1. Pour **50–100 mL** CuSO<sub>4</sub> solution into the jar.
2. Clip the **copper anode** to battery **(+)**; clip the **object** to battery **(-)**.
3. If using a resistor: put it in series so current must pass through it.
4. If using a multimeter for current: set to DC mA; wire it **in series** (not across the battery like a voltmeter).
5. Lower both electrodes into the solution so the plating area is submerged.
6. Keep electrodes **2–4 cm apart**; they must **not touch** (touching = short circuit).
7. Confirm bubbles or a slow color change starting within ~30–60 s. Read initial current if possible (**target ~10–40 mA** with a resistor; without a resistor current may be higher).

## Pre-plating record

| Item                                | Value |
|-------------------------------------|-------|
| Solution (0.5 M CuSO <sub>4</sub> ) |       |
| Approximate concentration if known  |       |
| Solution volume (mL)                |       |
| Solution color                      |       |
| Anode material                      |       |
| Cathode object                      |       |
| Electrode spacing (cm)              |       |
| Initial current (mA)                |       |
| Resistor used? (Y/N, Ω)             |       |

## Plating run (50–70 min)

1. Start the timer when the circuit is connected and current is flowing.
2. Every **1–2 minutes**, look at the object: color, shine, roughness, bubbles.
3. Typical good run: **3–8 minutes**. Stop sooner if the deposit turns black or powdery.

4. Disconnect the battery **before** removing the object.
5. Rinse gently under water; pat dry. Do not scrape the new copper.
6. Compare to an unplated control of the same object type.

### Observation log

| Time (min) | Appearance | Current (mA) | Notes |
|------------|------------|--------------|-------|
| 0          |            |              |       |
| 2          |            |              |       |
| 4          |            |              |       |
| 6          |            |              |       |
| 8          |            |              |       |
| Final      |            |              |       |

### What “good” looks like

| Appearance                   | Meaning                            | Action                                       |
|------------------------------|------------------------------------|----------------------------------------------|
| Bright salmon / copper color | Good plating                       | Continue or stop at planned time             |
| Dark brown / black powder    | Current too high                   | Stop; add resistor; re-sand; restart shorter |
| Patchy                       | Dirty surface or poor immersion    | Re-clean; ensure full immersion              |
| Heavy bubbles on object      | Competing H <sub>2</sub> evolution | Lower current / increase spacing             |

### Variable comparisons (70–80 min)

Groups test **one** variable (share results):

| Variable      | Options                                     | What to compare      |
|---------------|---------------------------------------------|----------------------|
| Plating time  | 2 vs. 8 min                                 | Thickness, color     |
| Current       | With vs. without 220–470 $\Omega$ resistor  | Smoothness           |
| Surface prep  | Clean vs. finger-greased                    | Adhesion, coverage   |
| Concentration | ~0.25 M vs. 0.5 M (if you dilute a portion) | Speed, color quality |
| Distance      | ~1 cm vs. ~4 cm electrode gap               | Uniformity, current  |

### Results table (class compilation)

| Group | Variable | Result | Best practice conclusion |
|-------|----------|--------|--------------------------|
|       |          |        |                          |



## Troubleshooting

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| Problem              | Cause                                   | Fix                                             |
|----------------------|-----------------------------------------|-------------------------------------------------|
| No plating           | Reversed polarity                       | Object must be on (–)                           |
| No plating           | Solution too weak / wrong bath          | Remake 0.5 M CuSO <sub>4</sub> ; check polarity |
| Black/powdery Cu     | Current too high                        | Add 100–500 Ω resistor; increase distance       |
| Patchy deposit       | Dirty / oily surface                    | Re-sand, rinse, no fingerprints                 |
| Anode not dissolving | Wrong anode (not copper)                | Use copper strip/wire                           |
| Battery gets hot     | Short circuit                           | Separate electrodes; check clips                |
| Very slow plating    | Low [Cu <sup>2+</sup> ] or bad contacts | Remake / concentrate bath; clean clips          |

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## Safety during experiment

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- Goggles on when working with solutions; gloves recommended
  - Do not ingest solutions; no mouth pipetting
  - Avoid short-circuiting the 9 V battery
  - Collect all copper-bearing liquid in the labeled waste container — **not** down the drain
  - Wash hands after the lab
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## Experiment status

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- [ ] 0.5 M CuSO<sub>4</sub> mixed and labeled
- [ ] Optimal plating time determined in pilot (record: \_\_\_ min)
- [ ] Resistor value chosen for ~10–40 mA (record: \_\_\_ Ω)
- [ ] Before/after photos for lecture use
- [ ] Variable comparison sheet printed for groups

## Syllabus Safety Notes

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Apply these across all sessions. Session-specific hazards are listed in each session folder.

### General rules

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1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M AgNO<sub>3</sub>** (Session 3) so chloride does not precipitate silver.

### Session-specific hazards

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| Session | Key hazard                               | Mitigation                                                                            |
|---------|------------------------------------------|---------------------------------------------------------------------------------------|
| 1       | Sharp nails/wires                        | Supervise handling; wash hands after metals                                           |
| 2       | Copper solutions                         | Goggles; no ingestion; collect waste                                                  |
| 3       | Silver nitrate                           | Gloves + goggles; stains skin/clothing; small volumes                                 |
| 4       | Gas evolution + H <sub>2</sub> balloon   | No saltwater; ammeter in series; instructor-only balloon/flame demo                   |
| 5       | Corrosion demos + electrolytic derusting | Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised |

### Hydrogen balloon / pop demo (Session 4, instructor-only)

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- Feed **cathode (H<sub>2</sub>) gas only** into a **small** balloon — do not mix with O<sub>2</sub>
- Inflate modestly; move away from the electrolysis bath and faces
- Long lighter at arm's length; clear area; extinguisher accessible
- **Never** ignite a sealed rigid container or a mixed H<sub>2</sub>/O<sub>2</sub> balloon
- Skip entirely if the venue forbids open flame